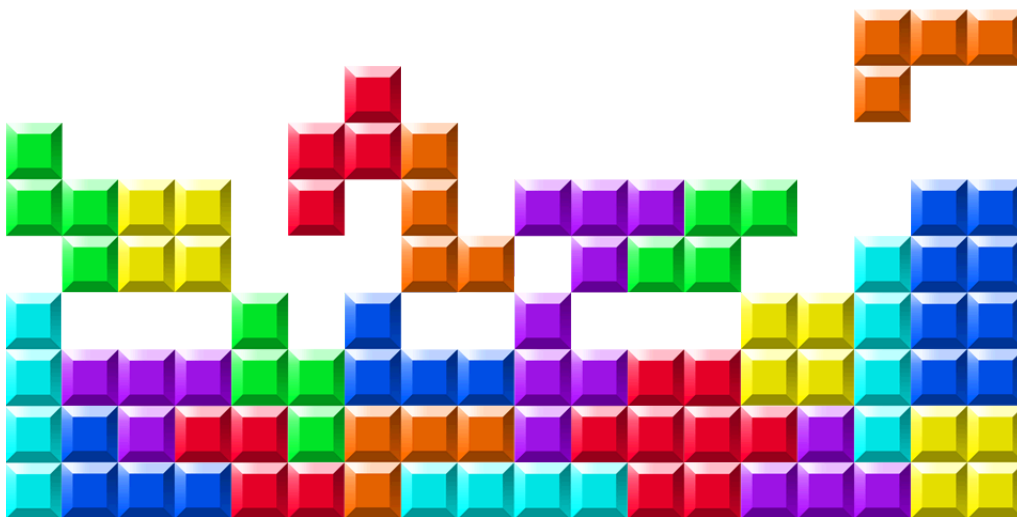

LINFO2275

Second project

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1 Introduction

2 Theoretical Background

2.1 Q-learning

Q-learning is a model-free reinforcement learning algorithm for learning optimal behaviour through interaction with an environment. It is model-free because it does not require a model of the environment, such as the transition probabilities or reward function. Instead, the agent learns from experience, by trying actions and observing the obtained rewards. In the standard formulation, the environment is modelled as a Markov decision process (MDP) defined by:

- a set of states $\mathcal{S}$,
- a set of actions $\mathcal{A}$,
- a transition probability distribution $p(s', r | s, a)$ that gives the probability of transitioning to state s' and receiving reward r after taking action a in state s ,
- a discount factor $\gamma \in [0, 1]$ that determines the importance of future rewards.

At time t , the agent observes a state S_t , selects an action A_t , receives a reward R_{t+1} , and moves to a new state S_{t+1} . The objective is to learn a policy that maximises the expected return, where the return is the discounted sum of future rewards:

$$G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}.$$

The agent learns by estimating action-values, also called Q-values. A Q-value represents how good it is to take a given action in a given state. More precisely, for a policy π , the action-value function is defined as

$$q_{\pi}(s, a) = \mathbb{E}_{\pi} [G_t | S_t = s, A_t = a].$$

The goal of Q-learning is to learn the optimal action-value function

$$q_*(s, a) = \max_{\pi} q_{\pi}(s, a).$$

The optimal Q-values satisfy the Bellman optimality equation:

$$q_*(s, a) = \sum_{s', r} p(s', r | s, a) \left[r + \gamma \max_{a'} q_*(s', a') \right].$$

In the tabular version of Q-learning, the agent stores one value $Q(s, a)$ for each state-action pair. After observing a transition

$$(S_t, A_t, R_{t+1}, S_{t+1}),$$

the Q-value is updated as follows:

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha \left[R_{t+1} + \gamma \max_{a'} Q(S_{t+1}, a') - Q(S_t, A_t) \right]$$

A common way to select actions during learning is the ε -greedy strategy. With probability $1 - \varepsilon$, the agent chooses the action with the highest current Q-value, while with probability ε , it chooses a random action.

3 Implementation

In this project, the Tetris environment is modeled as a Markov Decision Process (MDP) and solved using Deep Q-Networks (DQN). The implementation bridges the gap between raw game states and the reinforcement learning framework. The game and implementation of DQN are extensions of the work of Viet Nguyen [[tetrisVietnhGithub](#)].

3.1 Environment

Tetris is a game where blocs of different shapes and sizes appear at the top of a grid board and fall with time. The goal is to place as many blocks as possible without filling in the whole grid. If an entire row is filled when placing a block, that entire row disappears, and the player scores points. The highest number of rows that can be removed at once is 4, and doing so is called a *Tetris*.

Several different reward functions were explored for training the Q-Learning models. For the sake of thoroughness, up to 300,000 epochs per reward scheme were tested, and results for each model trained on 50,000, 100,000, 150,000, 200,000, 250,000, and 300,000 are shown in Figures 1, 2, and 3 in Section 4.2.

The three different reward schemes tested were:

- **Lines Flat:** The model receives +1.0 per line cleared, -2.0 for losing the game.
- **Lines Bonus:** Simulating original Tetris scoring behavior, the model receives +1.0 for clearing a single line, +3.0 for clearing two lines at the same time, +5.0 for clearing three lines at the same time, and +8.0 for clearing four lines at the same time. It also receives the same penalty of -2.0 for losing the game.
- **Per Block:** This model receives +1.0 per block placed and -2.0 for losing the game.

3.2 Action Space

Unlike traditional Tetris where moving pieces is possible while they fall, the agents have access to a *simplified action space*. An action a is defined as the tuple (x, r) , where x is the column where the piece should be dropped and r is the number of rotation of the piece. Since the grid is 10 units wide and a piece can be rotated at most 4 times, the action space contains 40 different actions.

3.3 Deep Q-Learning (DQN)

In complex environments like Tetris, the state space $\mathcal{S}$ is too large and therefore a traditional Q-value table would be very sparse and hard to learn from. Instead, Deep Q-Learning approximates $q_*(s, a)$ using a neural network $Q(s, a; \theta)$, where θ represents the weights of the network.

In this implementation, two distinct neural architectures are used: Neural Networks and Convolutional Neural Networks.

- **Convolutional Neural Network (CNN):** This architecture is used to process the 20×10 binary grid as an image. It utilizes three convolutional layers to extract spatial features and geometric patterns from the Tetris board, followed by dense layers to output the predicted Q-value.
- **Neural Network:** This model consists of three fully connected (linear) layers. It is designed to process a 4-dimensional *smart* feature state vector [lines, holes, bumpiness, height] (see

3.4). The layers use ReLU activation functions to introduce non-linearity, mapping the four features to a single scalar Q-value.

3.4 State Representation (S)

The state space is handled by the `get_state_properties` method in `tetris.py`, offering two distinct representations based on the chosen architecture.

The observation fed to the CNN is a 20×10 binary array where 1 represents the presence of a block, and 0 represents an empty space.

For the standard NN, the state representation is simplified to 4 heuristic features:

- **Lines Cleared:** number of lines cleared as a result of the given action.
- **Height:** Sum of the height of all columns.
- **Bumpiness:** Sum of the absolute differences in height between adjacent columns.
- **Holes:** Number of empty spaces with a filled space somewhere above it.

4 Experiments and Results

4.1 Comparison of state representations

To begin with, both the raw board CNN and heuristic NN models were given the same reward function for training.

The first obvious difference between the 2 was the amount of epochs needed for the models to understand the game.

JAKOB CREATE GRAPHICS TO COMPARE MODELS

In terms of behavior, the raw CNN model starts every game by placing blocks at random with no evident strategy. Only when it gets close to losing does it start placing blocks strategically, prolonging the game further.

The model fails to understand that the placement of the earlier pieces affects the score. Furthermore, it cannot take advantage of the exponential scoring of clearing multiple lines and does not seem to aim for Tetris.

On the other hand, the heuristic NN model is very efficient in its block placement from the very beginning. Frequently, it leaves an entire column empty so that it can fit a line piece to score a Tetris.

However, the model sometimes over-focuses on achieving Tetrises, and ends up losing the game before getting the line piece in.

4.2 Variations in given Rewards

Each epoch checkpoint was evaluated over 100 games with a maximum of 100,000 piece placements per game to prevent overly long games. During testing, the model always picked the action with the highest predicted Q-value with no random exploration.

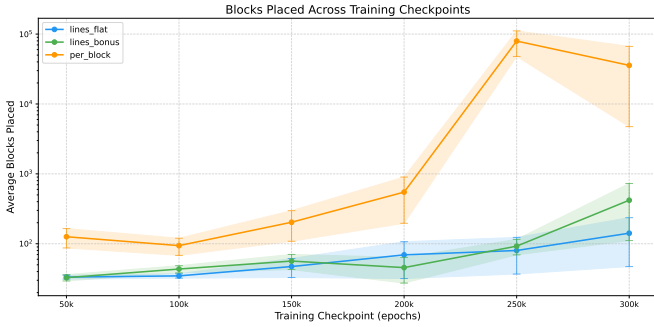


Figure 1: The average number of blocks placed by each reward model per game over 100 games at different training epochs.

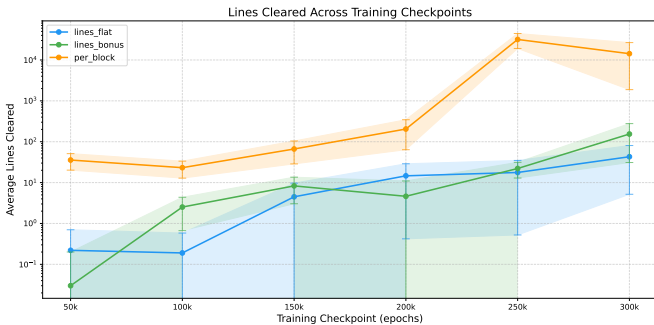


Figure 2: The average number of lines cleared by each reward model per game over 100 games at different training epochs.

As shown, the Per Block reward scheme produced by far the most resilient model from both the perspective of total blocks placed and lines cleared. The Flat Lines reward scheme seemingly produced the least consistent model, with many games resulting in little to no successful lines

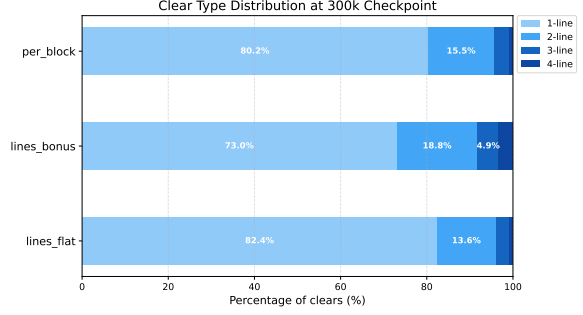


Figure 3: The distribution of number of lines cleared at same time by each reward model at different training epochs over 100 games.

cleared. As shown in Figure 3, the Lines Bonus scheme produced a model that favored multiple line clears at the same time, whereas Per Block displayed a stronger preference to clearing a single line at a time. This stands to reason, as the Lines Bonus was actively incentivized to clear as many lines at once as possible, whereas Per Block was more focused on surviving as long as possible.

One somewhat surprising result visible in Figures 1 and 2 is that some models actually perform worse at certain later checkpoints than earlier ones. This is seemingly a result of the replay buffer having a fixed size, so older experience gets overwritten over time. Therefore, if the model goes through a particularly bad patch, the buffer fills up with low-quality gameplay, the network trains on that, and performance drops further. This effect is made worse by the fact that by the time the model has reached 300,000 epochs, the value of epsilon is nearly zero, meaning there is essentially no random exploration to help the model recover from whatever suboptimal strategy it has settled into.